Molecular Immunology 135 (2021) 373–387

Contents lists available at ScienceDirect

Molecular Immunology

journal homepage: www.elsevier.com/locate/molimm

Recombinant FimH, a fimbrial tip adhesin of *Vibrio parahaemolyticus,* elicits mixed T helper cell response and confers protection against *Vibrio parahaemolyticus* challenge in murine model

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A R T I C L E I N F O

Keywords:
FimH
Vibrio parahaemolyticus
Vibriosis
Type 1 fimbriae
Vaccine
T helper cell response

A B S T R A C T

Vibrio parahaemolyticus causes vibriosis in wide range of marine organisms, and is responsible for food borne illnesses in humans through consumption of contaminated uncooked/partially cooked seafood. Continued and widespread antibiotics usage to increase the productivity has led to antibiotics resistance development. This has necessitated the need to develop alternative methods to control its infection. Use of safe and effective vaccines against the virulence factors not only protects from infection, it also minimizes antibiotic usage. The colonization of *V. parahaemolyticus* in the host and disease development requires several adhesins present on the cell surface, and thereby make them attractive vaccine candidates. *V. parahaemolyticus* produces extracellular type 1 fimbriae that have been shown to play a role in adhesion, biofilm formation and virulence. FimH is one of the minor components of the type 1 fimbriae occurring on its very tip. Being present on the cell surface, it is highly immunogenic, and can be targeted as a potential vaccine candidate. The present study describes the immunogenic and vaccine potential of recombinant *V. parahaemolyticus* FimH (r*Vp*FimH) expressed in *E. coli*. Immunization of BALB/c mice with the r*Vp*FimH elicited a strong mixed immune response, T-cell memory (evidenced by antibody isotyping, cytokine profiling and T-cell proliferation assay), and agglutination positive antibodies. FACS analysis and immunogold labeling showed that the polyclonal anti-r*Vp*FimH antibodies were able to recognize the FimH on *V. parahaemolyticus* cells. *In vivo* challenge of the r*Vp*FimH-immunized mice with $2\times LD_{50}$ dose of live bacteria showed one hundred percent survival. Thus, our findings clearly demonstrate the potential of FimH as an effective vaccine candidate against *V. parahaemolyticus*.

1. Introduction

Infectious diseases have remained a global threat to public health since the dawn of agriculture. The threat has become worse owing to climate change driven by global warming, development of antibiotic resistance, escalated import and export of foodstuff, migration, tourism and bioterrorism. Bacterial infections top the list of infectious diseases. Of these, emerging food borne gastrointestinal infections have raised grave concerns about food safety issues globally. The leading cause of foodborne "gastroenteritis" caused by *Vibrios*, also known as vibriosis, is *V. parahaemolyticus*, a halophilic, facultative-anaerobic, oxidase-positive, non-spore forming bacterium (Letchumanan et al., 2014; Farto et al., 2019). The bacterium is found as a free-living organism in marine and estuarine waters, sediments or associated with marine zooplanktons. These are one of the major contagions that infect humans as well as other organisms; as they are responsible for epizootic outbreak among marine fish population and infect human hosts that are sea-food consumers (Letchumanan et al., 2014). Approximately 45,000 of 80,000 vibriosis cases reported every year in the USA are caused due to *V. parahaemolyticus* infection (CDC report, 2021; https://www.cdc.gov/vibrio/index.html). Consumption of raw sea-food/water or contaminated food is primarily responsible for vibriosis as observed from most of the countries bordering oceans along coastal, estuarine, brackish and even fresh water worldwide (Esteves et al., 2015; Soto-Rodriguez et al., 2015; Ghenem et al., 2017). The hightened outbreak of *V. parahaemolyticus* mediated enteritis has been noticed during warm season, possibly due to the elevated sea surface temperature attributed to global warming, which is conducive to the bacterial proliferation

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https://doi.org/10.1016/j.molimm.2021.05.005
Received 28 January 2021; Received in revised form 21 April 2021; Accepted 9 May 2021
Available online 18 May 2021
0161-5890/

(Vezzulli et al., 2016).

V. parahaemolyticus infection in broad range of edible crustaceans, mainly shrimp results in acute necrotizing hepatopancreatitis (AHPND) (Soto-Rodriguez et al., 2015; Taha and Mohamed, 2020). Consumption of contaminated seafood results in gastrointestinal discomfort in humans and the infected individuals suffer from diarrhoea, cramps, nausea, vomiting, high fever, and dysentery with bloody or mucoid stools (Shimohata and Takahashi, 2010). The exposure of open wound to contaminated seafood and sea water also contributes to wound infection and also causes life-threatening septicemia in immuno-compromised patients (Johnson et al., 1984; Guillod et al., 2019). Thus, the bacterium extensively impacts both marine and human population, and severely affects the farmers involved in aquaculture/mariculture industry.

Widespread use of antibiotics as feed additives or for immersion bath for marine fishes meant for human consumption to control infection, and for the treatment of vibriosis has led to accumulation of antibiotics in the food chain, and development of antibiotics resistant strains, ultimately resulting in failure to control *V. parahaemolyticus* infection with antibiotics (Lee et al., 2018). Thus, emergence of multiple antibiotic resistant *V. parahaemolyticus* strains poses significant global threat towards seafood safety and public health. Further, *V. parahaemolyticus* establishes infection in the host through colonization and biofilm formation, which are orchestrated by several bacterial proteins. The biofilm enclosed bacteria are generally antibiotic and phagocytosis resistance, thus even if the strains by itself are not resistant to antibiotics, the antibiotics are rendered ineffective (Sharma et al., 2019).

Vaccination against several bacterial and viral infections has reduced the disease burden and death caused due to these infections, and has been proven to be the most effective approach in controlling the infectious diseases. Use of vaccines have also resulted in reduction in antibiotics usage and thereby reducing antimicrobial resistance development. Heat-killed cultures or live attenuated cultures are being currently used as commercial vaccine against vibriosis in white shrimp (Powell et al., 2011). These vaccines are not very efficacious as these formulations generate non-focussed immune response due to the presence of large number of bacterial antigens, in addition to posing the risk to persons involved in culturing the live bacteria. Thus, a subunit vaccine targetting a virulence factor that would intercept the establishment of infection in the host is desirable.

V. parahaemolyticus causes virulence through quorum sensing mediated cell-cell communication to form biofilm using an array of proteins involved at various stages of infection. Several outer membrane proteins involved in adhesion are flagella with swimming and swarming motility, pili and the metalloenzymes like enolase secreted on the extracellular surface play critical role during the initial stages of establishment of the bacterium in the host cells (Park et al., 2004; Okada et al., 2009; Wu et al., 2019).

The surface exposed virulent fimbrial/afimbrial adhesive organelles such as long whipped flagella, thinner hair like fimbriae or pili, toxins and outer membrane lipoproteins are assembled and exported to the surface as either end products or their integral components through a number of secretion pathways (Park et al., 2004; Okada et al., 2009; Costa et al., 2015).

The components of secretion system including several pili or fimbriae have been found to be highly virulent and have been identified as potential protective antigen against pathogenic bacteria responsible for a numerous human diseases (Sharma et al., 2011; Luiz et al., 2015; Forsyth et al., 2020). Attempts are being made to utilize different fimbrial adhesins as potential human vaccines towards urinary tract infection caused by uropathogenic *E. coli* (Duan et al., 2020; Forsyth et al., 2020 and references therein).

Type 1 fimbriae have been thoroughly investigated in uropathogens (Connell et al., 1996; Langermann et al., 1997; Hung et al., 2002; Klemm and Schembri, 2004; Wright et al., 2007). Approximately 100–500 fimbriae are present on the extracellular surface as long tubular, non-flagellar, peritrichous hair like adhesive structures. The fimbria comprises of heteropolymeric assemblies of major subunit FimA (2000 copies) and minor subunits FimF, FimG, FimH encoded by the *fimAICDFGHK* operon (Schwan, 2011). Presence of chaperone FimC and the usher subunit FimD is essential for the export and assembly of fimbrial subunits into a fimbria. The FimH of chaperone-usher pathway is the minor type 1 fimbrial tip adhesin. Crystal strucutre of FimC-FimH chaperone-adhesin complex from uropathogenic *E. coli* has been elucidated (Choudhury et al., 1999). FimH contains N-terminal mannose binding lectin/carbohydrate domain (1–156 amino acid), short intrapeptide loop (157–160 amino acid) to connect C-terminal pilin domain (161–279 aa) for integrating the tipped adhesin into the fimbrial structural assembly (Schembri et al., 2000; Munera et al., 2008; Busch and Waksman, 2012). Adhesion of type 1 fimbriae to the host cell is mediated through binding of the terminal adhesin FimH lectin domain to D-mannosylated proteins present on the eukaryotic host cell surface (Bouckaert et al., 2005; Tomašić et al., 2014). Interaction of type 1 fimbriae with the host cell is blocked by mannose or mannose derived FimH antagonists (Sokurenko et al., 1997; Klemm and Schembri, 2004; Mydock-McGrane et al., 2016).

A number of fimbrial adhesins have beern reported to be involved in the pathogenesis of different bacteria as they play an essential role in adhesion, and colonization. Few of these include FimH, PapG, Lpf1, F4, F5, F6, F18, F41,K99,CFA/I, S pilli and Dra adhesin of *E. coli* (Ascón et al., 1998; Klemm and Schembri, 2004 and references therein; Yang et al., 2011; Pereira et al., 2015), MrpH of *P. mirabilis* (Asadi et al., 2018); *Salmonella* fimbrial component SafD, FimA, FimA-mC3d2, and FimA-mC3d3 (Musa et al., 2014); FimA of *Porphyromonas gingivalis* (Choi et al., 2016); Fim2, Fim3, FimD of *Bordetella pertussis* (Guevara et al., 2016); *Bordetella bronchiseptica* Fim (Scheller et al., 2015). Immunization with some of these fimbrial proteins have been found to confer protective immunity against different bacteria namely *Yersinia pestis*, uropathogenic *E. coli*, *Salmonella enterica* (Chalton et al., 2006; Yang et al., 2011; Musa et al., 2014), albeit to different degrees.

It was noted that the *fimH* null mutants *of E. coli* showed relatively poor survival and neutrophil influx in comparison to the wild type *E. coli* in mouse urinary tract infection model, clearly indicating the role of FimH in bacterial establishment *in vivo* (Connell et al., 1996; for review please see Klemm and Schembri, 2004). Further, immunization of mice with enteropathogenic *E. coli* FimH, its fragments or its recombinant fusion with other surface proteins inhibited (>99 %) bacterial colonization in bladder (Langermann et al., 1997; Luna-Pineda et al., 2016). Polyclonal antiserum generated against the *E. coli* FimH inhibited binding of the bacterium with human bladder cells *in vitro* (Langermann et al., 1997). Immunization with *E. coli* FimCH fusion protein also conferred protection (~ 75 %) in monkeys against bacterial challenge (Langermann et al., 2000). Thus, vaccine potential of uropathogenic *E. coli* FimH is well established. FimH adhesin of type 1 fimbriae has also been found to be present in most of the virulent strains of *V. parahaemolyticus,* and is critical for bacterial adherence and invasion (Chao et al., 2010). The *V. parahaemolyticus* FimH can therefore effectively be used as vaccine candidates against the bacterium. The present study reports the immunogenic and protective potential of heterologously expressed recombinant FimH, a minor type 1 fimbrial tip adhesin against *V. parahaemolyticus* in mice model.

2. Materials and methods

2.1. Bacterial strains, vectors, chemicals

For recombinant cloning and expression studies, *Escherichia coli* DH5α and BL21 (λDE3) pLysS cells, respectively obtained from GIBCO-URL, USA were used. *V. parahaemolyticus* (MTCC#451) was procured from the Microbial Type Culture Collection (MTCC), Institute of Microbial Technology, Chandigarh, India. All reagents used in the study were of molecular biology grade and were obtained from Sigma Aldrich

Chemical Co., St. Louis, MO, USA, SD fine chemicals, Mumbai India or Sisco Research Laboratories, Mumbai, India. Components of bacterial culture growth medium were obtained from HiMedia Laboratories Pvt. Ltd., Mumbai, India.

2.2. Identification of the, fimH gene encoding fimbrial tip of the type 1 fimbriae from V. parahaemolyticus

The gene encoding the FimH of *V. parahaemolyticus* was identified by BLASTX analysis of *Escherichia coli* FimH against *Vibrionaceae* family. The gene and protein sequence of the identified accession encoding the *V. parahaemolyticus* FimH (*Vp*FimH) was extracted from the NCBI database. Expasy online tool was used to calculate molecular weight and pI of *Vp*FimH encoding sequence (www.expasy.org).

Amino acid sequences of FimH from different bacteria were retrieved from NCBI database. Constraint-based multiple Alignment Tool (COBALT), available at the National Center for Biotechnology Information, USA was used for multiple sequence allignment of the *Vp*FimH with other bacterial FimH (Papadopoulos and Agarwala, 2007). Evolutionary history was inferred from the phylogenetic tree constructed employing Neighbor-Joining method using MEGAX (Saitou and Nei, 1987).

The region encoding the mature *Vp*FimH was identified by analyzing the *Vp*FimH amino acid sequence for the presence of a signal peptide using SignalP 3.0 (www.cbs.dtu.dk /services/SignalP). A synthetic gene encoding the mature FimH of 851 bp (including the nucleotide sequences for *Nco*I and *Hind*III restriction cites at the 5' and 3' ends, respectively) was designed on the basis of sequence information of *V. parahaemolyticus fimH* gene [protein id, KKY42922.1; Geninfo identifier(GI), 82108904)]. The synthetic gene cloned into expression vector pET28a(+) at the *Nco*I and *Hind*III sites was procured from GenScript, USA and the recombinant plasmid designated as pET28.*VpfimH*. The recombinant protein produced from this construct would be of 294 amino acid residues, including 15 amino acid residues contributed from the vector (2 amino acid residues at the N-terminus and 13 residues at the C-terminus) as a result of cloning strategy.

2.3. Recombinant expression of histidine-tagged FimH of V. parahaemolyticus (rVpFimH)

Recombinant expression of the r*Vp*FimH was carried out essentially as described earlier (Karan et al., 2020). Briefly, a single colony of *E. coli* BL21 (λDE3) pLysS cells transformed with the plasmid pET28.*VpfimH* was used to inoculate the primary culture in LB containing Kanamycin (50 μg/mL) and allowed to grow overnight (37 °C with shaking at 200 rpm). A secondary culture inoculated with 1 % of primary culture was grown till O.D. reached 0.8 at 600 nm and induced with 1 mM IPTG for 6 h under similar conditions. The cell lysates from the uninduced and induced cultures were analyzed for expression of the r*Vp*FimH by SDS-PAGE (12 %) (Karan et al., 2020).

For localization of the expressed r*Vp*FimH, soluble and insoluble fractions of the sonicated cell lysate of the induced cells were prepared as described earlier (Yadav et al., 2016). The cell pellet from the cell culture was resuspended in sonication buffer (10 mM Tris-HCl, pH 8.0, 500 mM NaCl) and sonicated at 25 W, 40 amplitude for 20 min with 10 s pulse-on and 20 s-off time (MISONIX Ultrasonic Liquid Processors, USA). The sonicated lysate was centrifuged (12,000 rpm, Eppendorf microcentrifuge, USA) at 4 °C for 30 min to separate the soluble (supernatant) and insoluble (pellet) fractions. The fractions were analyzed by SDS-PAGE (12 %).

2.4. Purification of rVpFimH

SDS-PAGE analysis of subcellular fractions revealed the r*Vp*FimH to express as inclusion bodies (data not shown). Therefore, r*Vp*FimH was purified from the solubilized inclusion bodies fractions prepared by the method of Yadav et al. (2016). Purified inclusion bodies were resuspended in PENGU buffer (1 M urea, 5 % glycerol, 1 mM EDTA, 0.2 M sodium phosphate buffer, pH 8.0), followed by three washings and repeated suspension in PENGU buffer containing sodium deoxycholate (1 mg/mL). Solubilized inclusion bodies suspension was then incubated at room temperature (RT) for 15 min and centrifuged at 12,000 rpm (Eppendorf, USA, Model 5810 R, rotor FA-45−24-11) at 4 °C for 10 min. After washing the pellet once with PENGU buffer, it was solubilized in homogenization buffer (100 mM NaCl, 0.5 % Triton X-100, 0.1 % Sodium azide, 50 mM Tris-HCl, pH-8.0) and incubated at RT for 15 min. The final washing was performed with 10 mM Tris-HCl, pH 8.0 and further solubilised in 8 M urea prepared in 10 mM Tris-HCl, pH 8.0 for 30 min at RT. The r*Vp*FimH was purified from the solubilized inclusion bodies by Co^{2+}-NTA affinity chromatography as described earlier (Karan et al., 2020). The histidine-tagged r*Vp*FimH was allowed to bind with the Co^{2+}-NTA Sepharose pre-equilibrated with 8 M urea (in 10 mM Tris-HCl, pH 8.0) for 1 h at 4 °C. Repeated washing with 8 M urea-Tris-HCl solution was used for removing non-specifically bound proteins. The specifically bound r*Vp*FimH was eluted with 8 M urea-Tris-HCl solution containing 100 mM imidazole. Eluted fractions were analyzed by SDS-PAGE (12 %) and the fractions containing purified r*Vp*FimH were pooled and refolded using slow-pulsatile refolding method (5 % sucrose in 10 mM Tris-HCl, pH 8.0) as described by Mahmoodi et al. (2017). The refolded r*Vp*FimH protein was dialysed against 10 mM Tris-HCl, pH 8.0 containing 5 % glycerol, and concentrated by keeping the dialysed r*Vp*FimH over sucrose bed at 4 °C. The purified refolded r*Vp*FimH was immediately flash frozen in small aliquots for storage at −80 °C until further use. Protein concentration was estimated by BCA protein estimation kit (Bio-Rad, USA).

2.5. Generation of polyclonal sera against rVpFimH in BALB/c mice

Use of BALB/c mice (6 weeks, body weight 20 ± 2 g, n = 10 per group) for immunization and challenge studies was approved by the Institutional Animal Ethics Committee of the University (IAEC-JNU project code #08/2017). All animal experiments were performed as per the guidelines laid down by the IAEC. Animals were bled prior to immunization for collection of the control pre-immune (PI) serum. Purified and refolded r*Vp*FimH (15 μg; 1:1) or PBS emulsified in alum was administered intraperitoneally injected into each mouse (n = 10 per group). Subsequent booster doses with the same amount of protein or PBS were given on day 15 and 28. Mice were bled on day 14, 21 and 35 through retro-orbital plexus under light anaesthesia as described by Yadav et al. (2016). After incubation of the collected blood at 25 °C for 1 h, the sera collected by centrifugation at 7000 rpm (Eppendorf microcentrifuge, USA) at 4 °C for 7 min was stored at −20 °C (Yadav et al., 2016).

2.6. Determination of antigen-specific end point antibody titers and antibody isotypes titers by enzyme-linked immunosorbent assay (ELISA)

Antigen-specific end point titer of the anti-r*Vp*FimH antiserum was determined by ELISA as described by Yadav et al. (2016). ELISA plate were coated with purified r*Vp*FimH (500 ng/100 μl/well in 0.2 M carbonate-bicarbonate buffer, pH 9.2; 100 μl/well) and incubated overnight at 4 °C. The plates were washed with 1 × PBST (0.05 % Tween 20 in 1 × PBS) and after blocking the nonspecific sites with blocking buffer (2 % BSA in PBST; 200 μl/well), different dilutions of anti-r*Vp*FimH antisera (1:500, 1:1,000, 1:5,000, 1:10,000, 1:25,000, 1:50,000 and 1:100,000 in blocking buffer; 100 μl/well, in triplicate) were added to the wells and incubated at 37 °C for 1 h. After 3 washes with 1 × PBST (10 min each), horseradish peroxidase (HRP)-conjugated goat anti-mouse secondary antibody (100 μl/well, 1:5000 in 1 × PBST) was added to each well and incubated at 37 °C for 1 h. The wells were again washed with PBST and the color was developed with orthophenylene diamine prepared in citrate phosphate buffer [0.5 mg/mL, pH 5.5 and 30 % H_2O_2 (1 μL/mL); 100 μl/well] for 20 min at RT, followed by addition of 2 N H_2SO_4 (50 μl/well) to stop the reaction. The absorbance was measured at 450 nm in a microplate reader (BioTek Power Wave XS, USA).

ELISA was also used to determine the relative levels of different isotypes in the anti-r*Vp*FimH antisera drawn on different days post-immunization using isotype-specific ELISA kit (BD Pharmingen, USA), as directed by the manufacturer.

2.7. Immunoblot analysis

Immunoblotting of the uninduced and induced cell lysates, or purified r*Vp*FimH resolved on SDS-PAGE and transferred onto nitrocellulose membrane using anti-histidine monoclonal antibodies (Sigma Aldrich Chemical Co., St. Louis, USA) or anti-r*Vp*FimH antisera was carried out as described earlier (Yadav et al., 2016). Non-specific sites were blocked with 2 % BSA in PBST at 4 °C overnight. The blot was then incubated with anti-r*Vp*FimH antiserum (1 : 1000 in PBST 1 h, RT), followed by three PBST washes of 10 min each, and incubation with HRP-conjugated anti-mouse IgG (1:5000) for 1 h at RT. The blot was washed extensively with PBST, and the immunoreactive bands were detected using Pierce ECL Western blotting substrate (Thermo Scientific, USA). The image was acquired in a Biospectrum 500 Imaging system (UVP, Cambridge, UK).

2.8. In vitro splenocyte proliferation assay and analysis of T-cell response

On day 56 post-immunization, the splenocytes from the spleen of r*Vp*FimH- and PBS-immunized mice were isolated and subjected to T-cell proliferation assay as described by Yadav et al. (2016). After washing the splenocytes with 1 × RPMI-1640 containing 10 % fetal bovine serum medium (Gibco, Fisher Scientific UK), the cells were suspended in RBC lysis buffer (0.9 % NH_4Cl) and incubated at 4 °C for 5 min. Viable splenocytes were seeded in 96-well culture plates at a density of 1 × 10^5 cells/100 μl/well, stimulated with r*Vp*FimH (20 μg/mL) or PBS (control) and incubated at 37 °C for 24 h, 48 h and 72 h. The culture supernatants collected at different time intervals post-stimulation were stored at −80 °C for cytokine (IFN-γ and IL-4) ELISA. Splenocyte proliferation at different time post-stimulation was measured using XTT cell proliferation kit (Biological Industries, Israel) as per the direction of the manufacturer. The absorbance was measured at 450 nm (reference wavelength 570 nm) in an ELISA reader (BioTek Power Wave XS, USA).

The culture supernatant harvested at different time post-stimulation was analysed for IFN-γ and IL-4 levels using cytokine specific ELISA kit (Becton Dickinson, pharmingen) according to the manufacturer's instructions. The culture supernatants (collected 72 h post-stimulation) from the splenocytes isolated from the r*Vp*FimH-immunized mice, and stimulated with the r*Vp*FimH or PBS were analyzed for global cytokine response using Proteome Profiler antibody array (R&D Systems, USA) as per the protocol provided by the manufacturer.

2.9. In vitro agglutination assay

Analysis of anti-r*Vp*FimH mediated agglutination of *V. parahaemolyticus* cells was carried by the method of Karan et al. (2021). For this, the *V. parahaemolyticus* cells at mid-log phase were collected by centrifugation and washed with 1 × PBS. Agglutination reaction was carried by incubating the cells in different dilutions of the anti-r*Vp*FimH or pre-immune antisera [1 × 10^8 colony forming units (CFU) cells suspended in antisera dilutions of 1:100, 1:200, 1:300, 1:500 in 1 mL of 1 × PBS] at 37 °C for 2 h. The cells were again collected by centrifugation, washed once with 1 × PBS and resuspended in 100 μL of 1 × PBS. The glass slide smear of the cells was air-dried, heat-fixed, and stained with methylene blue solution (1 % methylene blue dissolved in 95 % ethanol and 0.01 % KOH). The smear was washed with Milli-Q water and the images were acquired using bright-field microscope (Model Eclipse TE2000S, Nikon, USA).

To check the specificity of agglutination of the bacterial cells by the anti-r*Vp*FimH antiserum, the antiserum was incubated with purified r*Vp*FimH (1 μg/μL of 1:300 antisera dilution) for 1 h at 37 °C prior to addition to the bacterial cells.

Agglutination of piliated *E. coli* has been reported to be Mannose-sensitive (van der Bosch et al., 1980). The FimH has a lectin binding domain and has high affinity to mannose. Therefore, presence of mannose will abolish or reduce the binding of anti-FimH antibodies present in the antiserum and accordingly reduce or abolish agglutination. Therefore, one agglutination reaction was set with *V. parahaemolyticus* cells (mid-log phase, 1 × 10^8 CFU), pre-incubated with 50 μM mannose at RT for 10 min, prior to the addition of anti--r*Vp*FimH antiserum (1:200 in 1 × PBS). The extent of agglutiantion was visualized under microscope as described earlier.

2.10. Recognition of V. parahaemolyticus FimH by anti-rVpFimH antibodies

The specific reaction of anti-r*Vp*FimH antibodies present in the antisera with native FimH of *V. parahaemolyticus* was investigated by flow cytometry and immunoelectron microscopy.

2.10.1. Flow cytometry

The cross-reactivity of anti-r*Vp*FimH antisera generated against the r*Vp*FimH with the FimH present on the cell surface of *V. parahaemolyticus* was confirmed by fluorescence-activated cell sorting (FACS) as described earlier by Karan et al. (2021). *V. parahaemolyticus* cells (1 × 10^6 CFU) at mid-log phase culture were collected by centrifugation (4000×g, 10 min) and washed with chilled 1 × PBS. Anti--r*Vp*FimH antiserum or pre-immune serum (1:200 in 1 × PBS) was added to the cells, incubated for 1 h at 4 °C, followed by centrifugation and washing with 1 × PBS. Fluorescein isothiocyanate (FITC)-conjugated AffiniPure Goat Anti-Mouse IgG (1:200 dilution prepared in 1 mL of 1 × PBS) was then added to the cells and incubated for 30 min at 4 °C. FACS analysis (excitation 495 nm, emission 519 nm) was carried out to assess the binding of anti-r*Vp*FimH antibodies with the native *V. parahaemolyticus* FimH using FACS Calibur™ (Becton Dickinson Immunocytometry System, San Jose, CA). The fluorescence data were acquired and analyzed using Flow Jo 10.7.1 software.

2.10.2. Immunoelectron microscopy

Immunogold labeling of *V. parahaemolyticus* using anti-r*Vp*FimH antibodies was performed by the method of Karan et al. (2021) to further confirm the interaction of anti-r*Vp*FimH antibodies with the *V. parahaemolyticus* FimH. Mid-log phase *V. parahaemolyticus* cells were washed with 1 × PBS at 4 °C and incubated in TEM fixative (4 % paraformaldehyde and 0.5 % glutaraldehyde in 1 × PBS) for 60 min at RT once and again for 10 min at 4 °C. Quenching solution (100 mM ammonium chloride in 1 × PBS) was added to the fixed cells and allowed to incubate for 10 min at 4 °C. After washing the cells with 1 × PBS at 4 °C twice, the cells were encased in molten agar (2.5 % agar in 1 × PBS) using London Resin white resin and were subjected to sectioning (100 nm thin sections) on Nickel grids using Leica Ultracut (Leica, Germany). The cells were then processed for immunogold labeling using incubation with primary antibody (Anti-r*Vp*FimH or preimmune serum at a dilution 1:20 in 1 × PBS containing 2 % BSA), followed by three PBST washes of 10 min each. Colloidal (10 nm) gold-labelled anti-mouse antibody was then added to the grid and incubated at RT for 60 min. After several washes with PBS, the grids were stained with saturated solution of uranyl acetate solution (2 %) at RT for 30 s, and visualized under transmission electron microscope (Tecnai F20 TEM, USA) at the National Institute of Immunology, New Delhi.

2.11. In vitro adhesion and invasion assay of V. parahaemolyticus

V. parahaemolyticus infects gastrointestinal tract and therefore *in vitro* adhesion and intracellular survival assay using mammalian colon cancer cell line HCT116 (ATCC#CCL247, National Centre for Cell Science, Pune, India) was performed, essentially as described earlier (Luna-Pineda et al., 2016). The HCT116 cells were cultured and maintained in DMEM with 10 % fetal bovine serum medium at 37 °C in 5 % CO_2 incubator in culture flask. For adhesion and invasion assay, the cells seeded in a 24-well culture plate (1 × 10^5 cells/1000 μl/well) were grown in the 5 % CO_2 atmosphere overnight. Next day, 1 × 10^7 colony forming units (CFU) of *V. parahaemolyticus* cells (late log phase) were mixed with either the anti-r*Vp*FimH (1:50, 1:100 in 1 mL 1 × PBS) or pre-immune antisera (1:50 in 1 mL 1 × PBS) and incubated at 37 °C for 2 h. Post-incubation, *V. parahaemolyticus* cells collected by centrifugation (4300×g, 10 min) were washed with 1 × PBS, resuspended in DMEM containing 10 % fetal bovine serum and and added to HCT116 cells monolayer (multiplicity of infection, 1:100) for 2 h at 37 °C in 5 % CO_2 atmosphere. Non-adhered cells were removed by thrice washing with DMEM. After removing the medium carefully, the cells were washed with DMEM to remove non-adhered bacterial cells. The HCT116 cells were then lysed using [100 μl of lysis buffer (0.1 % Triton X-100 in 1 × PBS) for 10 min at RT, followed by addition of 900 μl LB broth and thorough mixing. Different dilutions of *V. parahaemolyticus* cell suspension in LB broth was spread plated on LB agar and incubated overnight at 37 °C. The colonies on the LB agar plate were counted for determining adhered and intracellular *V. parahaemolyticus* cells.

2.12. *In vivo* challenge studies

The CFU corresponding to LD_{50} dose of the *V. parahaemolyticus* was determined by the method of Reed and Muench (1938) by challenging unimmunized mice with different CFU of bacterial cells. The number of *V. parahaemolyticus* cells (~2.5 × 10^8 CFUs) that resulted in the death of 50 % of animals within 24 h was taken as the LD_{50} dose. Therefore, to attain 100 % death of the control mice in challenge studies, twice the LD_{50} (2×LD_{50}) dose of *V. parahaemolyticus* (~5 × 10^8 CFUs) was administered to immunized mice.

Neutralizing ability of the anti-r*Vp*FimH antiserum was assessed by passive challenge of unimmunized normal BALB/c mice (n = 10). The mice were challenged intraperitoneally with 2×LD_{50} dose of *V. parahaemolyticus* cells resuspended in anti-r*Vp*FimH antisera (50 μL) and incubated at 37 °C for 1 h. Control group was challenged with the same amount of cells pre-incubated with neat pre-immune serum. Health and survival of the mice were monitored for 10 days.

A separate set of the r*Vp*FimH-immunized and PBS-immunized mice, given a 3rd booster on day 60 was used for active challenge study. The immunized mice were challenged with 2× LD_{50} dose of *V. parahaemolyticus* (IP) a week after the 3rd booster and monitored for health and survival for 10 days.

2.13. Statistical analysis

The data represent mean ± SD (standard deviation) of three independent experiments, each performed in triplicate. Paired Student's *t*-test was performed to determine statistical significance (*p* value).

3. Results

3.1. *V. parahaemolyticus fimH* gene sequence

The sequence of type 1 fimbrial adhesin FimH sequence from *V. parahaemolyticus* (*Vp*FimH) with protein id as KKY42922.1 (Geninfo identifier # GI: 82108904) was identified by BLASTX analysis of *E. coli fimH* gene sequence (Accession no. JX847135.1) against *V. parahaemlyticus* genome sequence (GenBank: LCUL01000008.1). The encoded protein of 301 residues gives rise to a mature *Vp*FimH of 279 amino acid residues, after the removal of signal peptide of 22 amino acid residues.

Multiple sequence alignment of the *Vp*FimH amino acid sequence with the FimH of other bacteria showed varying degree of percentage identity ranging between 24.39 % to 99.67 % (Supplementary Table S1). The *Vp*FimH showed maximum identity (99.67 %) with *Citrobacter amalonaticus* minimum percentage identity (24.39 %) with *Enterobacter roggenkampii*.

Phylogenetic tree, with the sum of branch length of 4.77619522, derived from the multiple sequence alignment of the *Vp*FimH amino acid sequence with that of other bacteria (Supplementary Fig. S1) showed the *Vp*FimH to be closely related with the FimH of *Citrobacter amalonaticus, Citrobacter farmeri, Klebsiella, Shigella, Escherichia, Raoultella* and *Erwinia.*

3.2. Expression and purification of recombinant FimH of *V. parahaemolyticus* (r*Vp*FimH)

Induction of expression of the r*Vp*FimH in *E. coli* BL21 (λDE3)pLysS cells transformed with the pET28.*VpfimH* with IPTG did not appear to induce a noticeable increase in the expression as a very faint band at the expected size of ~32 kDa, with a slightly greater intensity was observed in the induced cell lysates (Fig. 1A, lane I), when compared to uninduced cell lysates (Fig. 1A, lane U). However, sonication of the induced cell lysate showed exclusive presence of the r*Vp*FimH in the insoluble fraction (Fig. 1B, lane P). Negligible expression was detected in the soluble fraction (Fig. 1B, lane S). Western blot analysis of the induced cell lystes and the soluble and insoluble fraction using anti-6×Histidine tag monoclonal antibodies further confirmed the authenticity of the recombinant protein to be histidine tagged r*Vp*FimH (Fig. 1C and D). A single band at the expected size of the histidine-tagged r*Vp*FimH was detected in the induced cell lysate (Fig. 1C, lane I) and the insoluble fraction (Fig. 1D, lane P) of the induced cell lysate.

Purification of the r*Vp*FimH protein from the solubilized inclusion bodies using metal affinity chromatography resulted in a highly purified r*Vp*FimH to a near homogeneity of ~ 99 % (Fig. 1E, lanes E1-E3) and a yield of ~43.2 mg/l at the shake flask level. MALDI-TOF-MS of the refolded r*Vp*FimH further confirmed the purified protein to be type 1 fimbrial protein FimH of *V. parahaemolyticus* (Supplementary Fig. S2).

3.3. Analysis of immune response

3.3.1. Immunogenic potential of the r*Vp*FimH

Antigen specific end point titer of the anti-r*Vp*FimH antisera were determined to be very high (>~1:100000) (Fig. 2A). A significant increase in the immunoglobulin IgG levels ($p \leq 0.001$-0.005) was observed a week after the administration of first booster dose. However, additional booster administration did not bring about any further increase in the IgG levels in the anti-r*Vp*FimH antiserum.

The antigenicity of the r*Vp*FimH was further confirmed by Western blot analysis using anti-r*Vp*FimH antiserum (Fig. 2B). An immunoreactive band at the expected size of the r*Vp*FimH (~32 kDa) confirms the antigenicity of the r*Vp*FimH.

3.3.2. Immune response analysis by antibody isotyping

The immunoglobulin isotyping ELISA of the anti-r*Vp*FimH antisera showed the switch of immunoglobulin expression and isotypes differentiation in the anti-r*Vp*FimH antisera collected on day 14, 21 and 35 post-immunization (Fig. 3). Significant higher levels ($p \leq 0.001$-$p \leq 0.005$) of all the isotypes *i.e.*IgG1, IgG2a, IgG2b and IgA were observed on day 14. While the IgG1, IgG2b and IgA levels remained constant in the antisera drawn after two booster doses, the IgG2a levels decline slightly, but still remained significantly higher in comparison to that of pre-immune serum. The ratio of IgG1:IgG2a (~2.22, ~5.29, ~6.87) and IgG1/IgG2b (~1.69, ~1.99, ~2.53) was determined in the antisera drawn on day 14, 21 and 35, respectively.

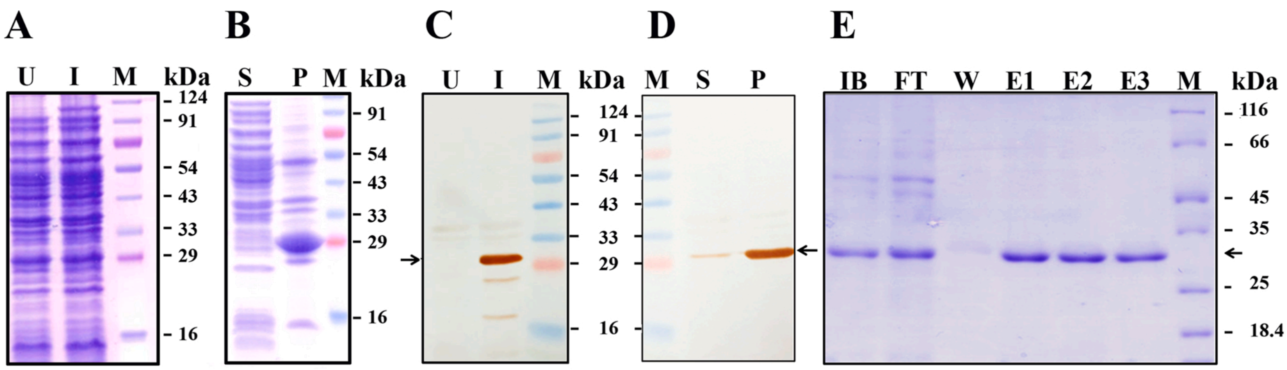

Fig. 1. Expression analysis of recombinant FimH protein in *E. coli* BL21 (λDE3) pLysS. **(A)** SDS-PAGE analysis of the total cell lysates of the uninduced (U) and induced (I) *E. coli* BL21 (λDE3) pLysS harboring the pET28.*VpfimH* **(B)** Lanes "S" and "P" represent soluble and insoluble fractions of the induced cell lysates of *E. coli* BL21 (λDE3) pLysS harboring the pET28.*VpfimH* analyzed by SDS-PAGE (12 %). **(C and D)** Western blot analysis.The uninduced (U) and induced (I) cell lysates (panel C), and soluble (S) and insoluble (P) fractions (panel D) of the induced cell lysates of *E. coli* BL21(λDE3) pLysS cells harboring pET28.*VpfimH*, resolved on SDS-PAGE were immunoblotted using anti-His antibody (1:3000 in 1 × PBS containing 0.05 % Tween-20). **(E)** Purification profile of the recombinant FimH of *V. parahaemolyticus* (r*Vp*FimH). SDS-PAGE (12 %) analysis of r*Vp*FimH purified from the urea solubilized inclusion bodies fraction (IB) of the induced cell lysate. Lanes IB, FT, W, IB, and E1-E3 represent insoluble fraction solubilized in 8 M urea, flow through, wash fraction, and fractions eluted with buffer containing 100 mM, 200 mM, 250 mM Imidazole. "M" indicates protein molecular weight (kDa) markers. Arrow in all the panels points to the expressed recombinant protein.

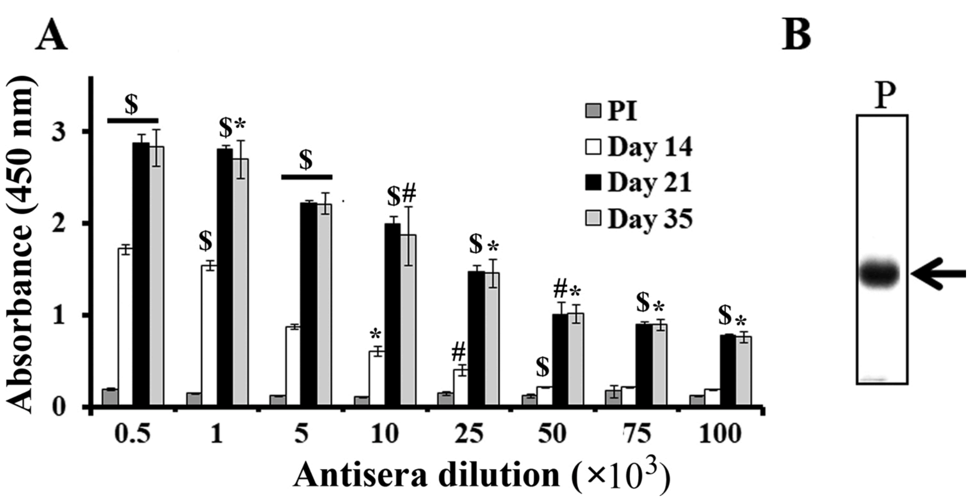

Fig. 2. Antigenicity analysis of the r*Vp*FimH. **(A)** Antibody end-point titer determination. ELISA was performed with Antisera from the r*Vp*FimH-immunized mice for determination of the r*Vp*FimH-specific antibody titer. Different dilutions of the primary antibody (in 2 % BSA in 1 × PBS containing 0.05 % Tween-20) were used. Dilution of the secondary antibody was kept at 1:5000 dilution. Pre-immune (PI) serum was included for baseline control. Significant increase in absorbance obtained with anti-r*Vp*FimH antisera is calculated with that obtained with PI are shown ($, $p \leq 0.001$; *, $p \leq 0.005$; #, $p \leq 0.05$). **(B)** Western blot analysis. The purified r*Vp*FimH was immunoblotted with anti-r*Vp*FimH antiserum (1:2000 in 1 × PBS containing 0.05 % Tween-20). A single band at the expected size of the purified protein (P) at the (~32 kDa) is detected..

3.3.3. *In vitro T-cell response by the rVpFimH*

Immunization with the r*Vp*FimH was able to generate T cell memory as established by splenocytes proliferation assay. Significant increase in splenocytes proliferation was observed in the r*Vp*FimH-stimulated splenocytes isolated from the r*Vp*FimH-immunized mice at 24 h ($p \leq 0.001$), 48 h ($p \leq 0.005$) and 72 h ($p \leq 0.001$) post-stimulation as compared with the r*Vp*FimH-stimulated splenocytes isolated from the PBS-immunized mice and PBS-stimulated splenocytes isolated from these two groups (Fig. 4A). Proliferation index (A_{450} stimulated /A_{450} unstimulated) was determined to be as ~1.51, ~1.46 and ~1.93 for the r*Vp*FimH immunized mice at 24 h, 48 h and 72 h post-stimulation, respectively. The culture supernatants collected from the r*Vp*FimH stimulated splenocytes isolated from the protein immunized mice showed a significant enhancement in the IFN-γ ($p \leq 0.001$) and IL-4 ($p \leq 0.001$) levels at all the three time point (Fig. 4B and C). Stimulation of splenocytes with r*Vp*FimH resulted in a 3 fold increase in the the levels of both IFN-γ and IL-4 was noted at 24 h in comparison to that stimulated with PBS. A further increase in both the cytokines was observed at 48 h post-stimulation. Though IFN-γ levels declined slightly at 72 h in comparison to that at 48 h, the levels were still significantly higher than the corresponding PBS controls. IL-4 levels showed further increase at 72 h post-stimulation, again confirming Th2-biased response. In comparision to the respective cytokine levels from the r*Vp*FimH-stimulated splenocytes isolated from PBS-immunized mice, the increase was determined to be ~10 and ~9 folds at 48 h and ~ 5 fold and ~12.75 folds at 72 h for IFN-γ and IL-4 levels, respectivley. No significant change in the levels of these cytokines with time was observed in the supernatant from the r*Vp*FimH- stimulated splenocytes isolated from the PBS immunized mice.

In the global cytokine array analysis, out of the 40 antibodies arrayed for different cytokines/chemokines, significantly increased levels of 27 cytokines/chemokines were observed in the culture supernatants of the splenocytes isolated from the r*Vp*FimH-immunized mice stimulated with r*Vp*FimH, in comparison to the culture supernatant of r*Vp*FimH-immunized mice stimulated with PBS (Fig. 5). Highest difference at the significance level of $p \leq 0.001$ was observed in the CCL3, CXCL2, IL-17, IL-2, CCL5, sICAM-1, KC, TNF-α, IL-5, IL-13, IL-10, IL-3, IL-4, CCL4, IL-1α, IL-1β. Next highest level of change at $p \leq 0.005$ was noted in the levels of IP-10, GM-CSF, G-CSF, M-CSF, IL-6, JE and IL-1ra, whereas IFN-γ, MIG, I-309, IL-7 showed an increase at significance level of only $p \leq 0.05$. These data further confirm generation of mixed immune response upon r*Vp*FimH immunization. Based on the secreted cytokines and their abilities to assist other immune cells, the populations of Th CD4+ cells were identified as Th1, Th2 and also Th17 cells.

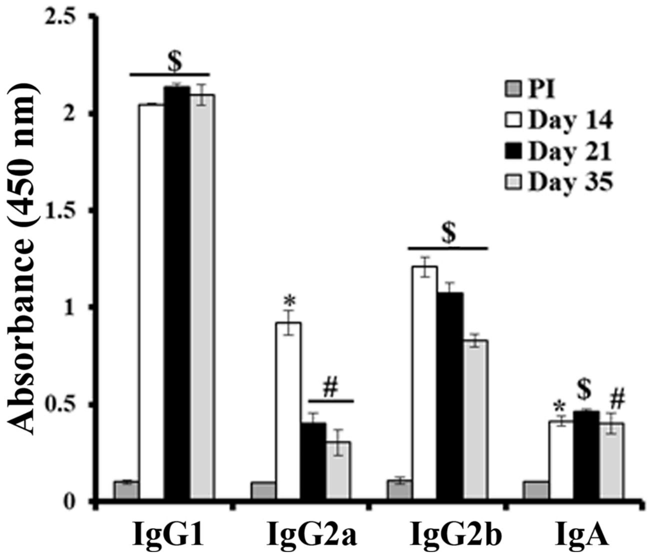

Fig. 3. Determination of immunoglobulin isotypes of the anti-r*Vp*FimH antisera. Polyclonal antibody isotypes in the immunized mice sera were determined by ELISA. ELISA was performed to determine levels of different antibody isotypes in the anti-r*Vp*FimH antisera and pre-immune (PI) collected on different days post-immunization were determined using isotype-specific antibodies Statistical significance (($, $p \leq 0.001$; *, $p \leq 0.005$; #, $p \leq 0.05$) is determined with respect to the levels of respective isotype in pre-immune serum..

3.4. Neutralization potential of the anti-rVpFimH antisera and specificity of anti-rVpFmH antibodies interaction with V. parahaemolyticus FimH

Incubation of *V. parahaemolyticus* with the varying dilutions of anti-r*Vp*FimH antisera showed positive agglutination, whereas no clumping of the cells was observed with pre-immune sera (Fig. 6A). With the increase in antisera dilution, decreased agglutination of the bacterial cells was observed. Specificity of the agglutination by the anti-r*Vp*FimH antisera was established by inhibition of the clumping of *V. parahaemolyticus* cells, when the anti-r*Vp*FimH antiserum was pre-incubated with the r*Vp*FimH, prior to addition to *V. parahaemolyticus* cells (Fig. 6B, upper panels).

The native FimH present on the bacterial cells interacts with the host cell through mannosylated receptor. Therefore, the anti-r*Vp*FimH antibody is expected to bind to FimH resulting in blocking of mannose binding and *vice versa* (*i.e.* mannose would inhibit agglutination ability of the anti-r*Vp*FimH antiserum). As shown in Fig. 6B (lower panels), *V. parahaemolyticus* cells pre-incubated with mannose did not show any clumping and individual cells could be clearly seen. These data confirm that the anti-r*Vp*FimH antibodies indeed cross-reacted with the lectin binding domain of FimH present on the bacterial cell surface.

To establish that the agglutination was due to specific interaction of the anti-r*Vp*FimH antibodies present in the antisera with the FimH present on the bacterial cell surface, cross-reactivity analysis of anti-r*Vp*FimH antisera with the FimH present on the extracellular surface of *V. parahaemolyticus* cells was confirmed by gold immunolabelling using TEM (Fig. 7). Specific dark spots could be detected at the fimbrial tips on the bacterial cell surface when the cells were treated with anti-r*Vp*FimH antiserum. No such spots were evident in the TEM images of the *V. parahaemolyticus* cells incubated with pre-immune serum.

The interaction of the anti-r*Vp*FimH antibodies with the *V. parahaemolyticus* FimH was further confirmed by FACS using FITC-conjugated anti-mouse IgG secondary antibody (Fig. 8). The cells incubated with anti-r*Vp*FimH antiserum showed significantly higher (~4.6 fold greater; $p \leq 0.001$) fluorescent cell population (44.77±4.72 %) in comparison to that incubated with pre-immune serum (9.72±0.41 %).

Specific interaction of the anti-r*Vp*FimH antibodies with the FimH on

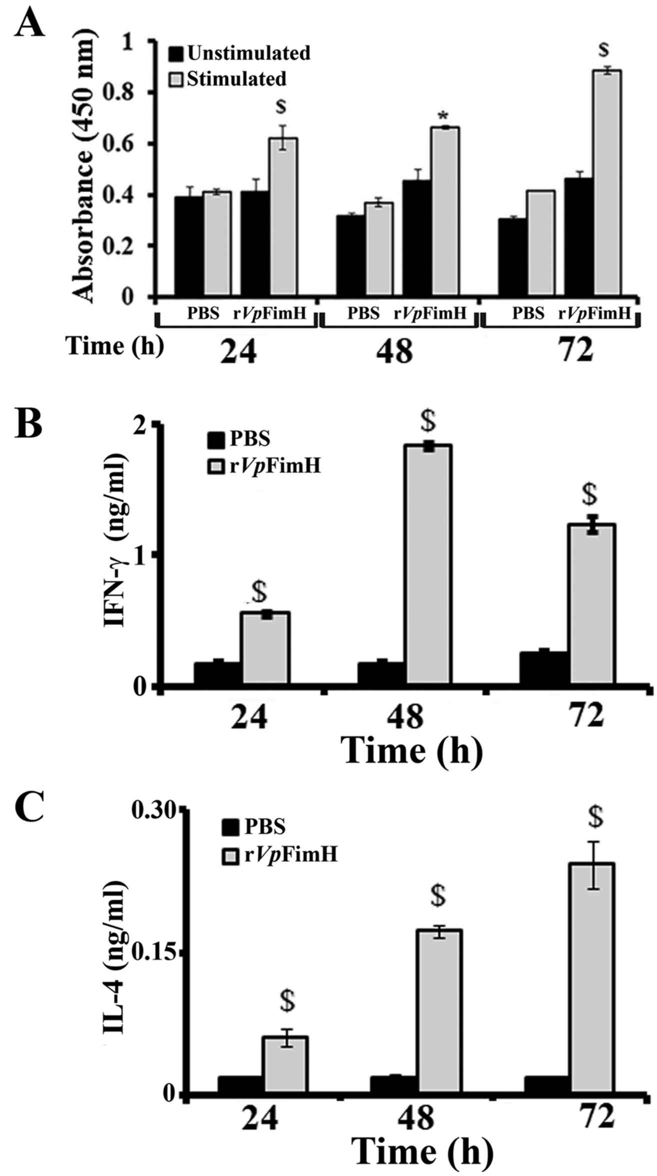

Fig. 4. *In vitro* splenocytes proliferation assay. **(A)** Stimulation of splenocyte proliferation by the r*Vp*FimH. Antigen-specific proliferative response of splenocytes isolated from the PBS- and r*Vp*FimH-immunized mice was determined after stimulation with the r*Vp*FimH (stimulated). Cells stimulated with the corresponding volume of the PBS (unstimulated) was included as control. Proliferation of the cells incubated in a humidified CO_2 (5 %) incubator for different time periods (24 h, 48 h and 72 h) was assayed using XTT cell proliferation assay. Significance ($,$p \leq 0.001$;*, $p \leq 0.005$) levels of changes in the r*Vp*FimH-stimulated splenocytes were calculated with respect PBS-stimulated splenocytes. **(B-C)** IFN-γ and IL-4 production by the stimulated splenocytes of the immunized mice. The two cytokines were measured in the culture supernatants of the splenocytes isolated from the r*Vp*FimH-immunized mice, collected at different time points post-stimulation with the rVpFimH- or PBS using cytokine specific ELISA. Cytokine specific antibodies were used at 1:250 diltuion whereas and detection antibodies for IFN-γ (panel B) and IL-4 (panel C) were used at 1:500 and 1:250 dilution, respectively The *p*- values ($, $p \leq 0.001$; *, $p \leq 0.005$) indicate significance levels of change with respect to the levels of respective cytokine in PBS-stimulated splenocytes.

V. parahaemolyticus cell surface was also studied by analyzing its effect on adherence and invasion of *V. parahaemolyticus* cells on mammalian colorectal carcinoma cells (HCT116). For this, the bacterial cells were incubated with the antiserum prior to addition to HCT116 cells (Fig. 9). A significant decline ($p \leq 0.001$) in the average number of adhered and

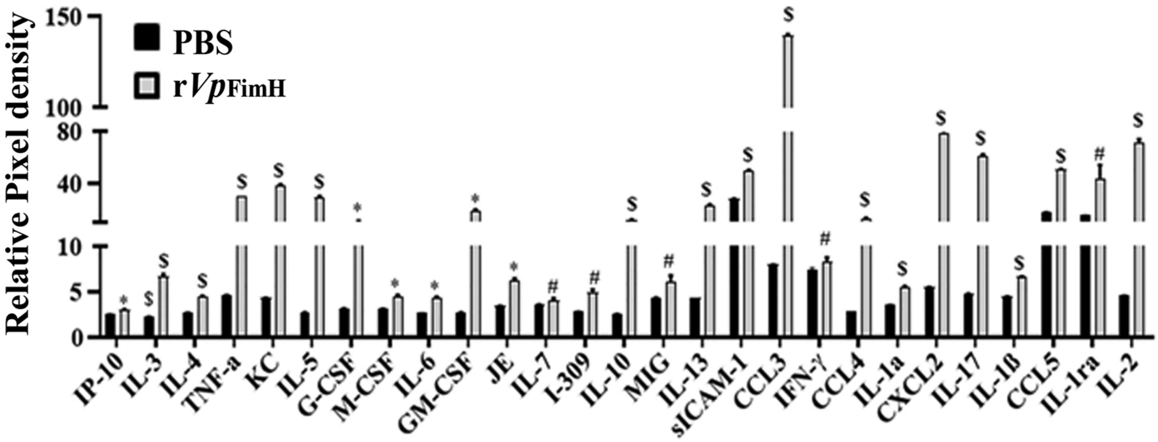

Fig. 5. Global cytokine respohnse analysis. Cytokine array kit (R & D systems, USA) was used to determine the relative levels of different cytokines in the culture supernatants collected at 72 h from the control PBS-stimulated and r*Vp*FimH-stimulated splenocytes, isolated from the r*Vp*FimH-immunized mice (Supplementary Fig. S2). The figures depicts the relative pixel density of the cytokines/chemokines which showed significant change with respect to PBS-stimulated splenocytes. $, $p \leq 0.001$; *, $p \leq 0.005$; #, $p \leq 0.05$.

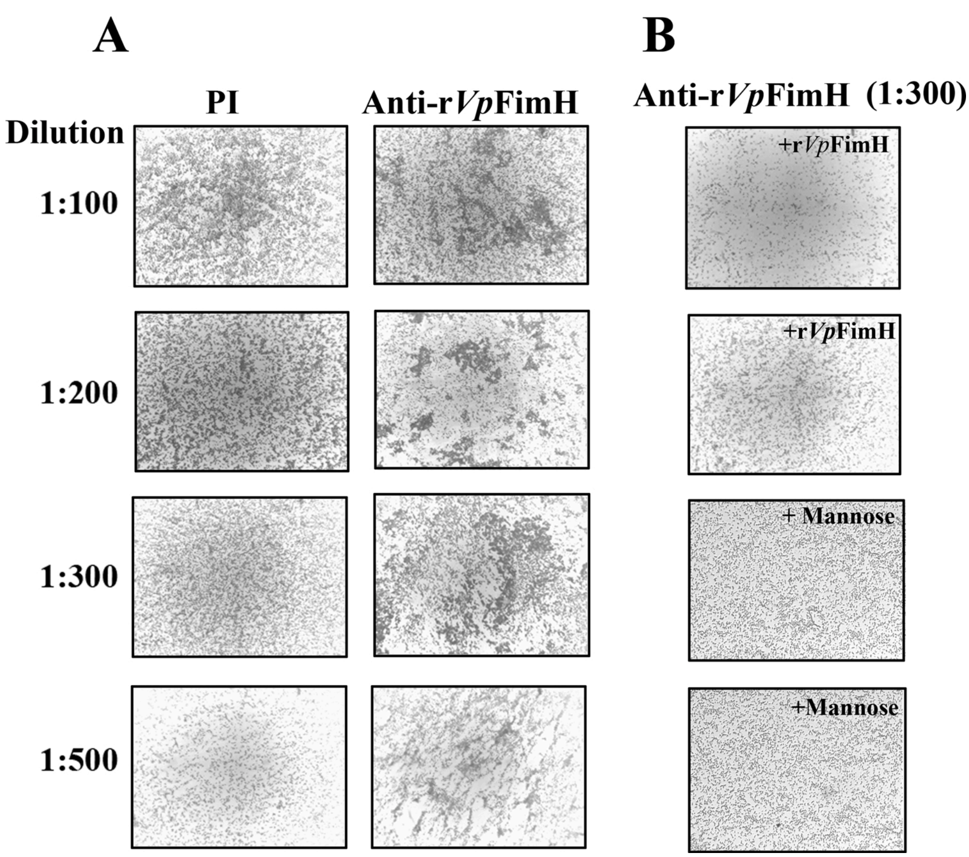

Fig. 6. *In vitro* analysis of inhibition of agglutination by the anti-r*Vp*FimH antisera. **(A)** *V. parahaemolyticus* (MTCC#451; mid-log phase cells, 1×10^8 CFU),resuspended in different dilutions of anti-r*Vp*FimH antisera (in 1 × PBS) were incubated at 37 °C for 2 h. **(B)** Anti-r*Vp*FimH antisera (1:300 dilution) was pre-incubated with the r*Vp*FimH (1 μg/ μl of anti-sera at 1:300 dilution) and 50 μM D-Mannose (in duplicate) prior to addition to *V. parahaemolyticus* cells followed by incubation for 2 h at 37 °C. Agglutination of bacteria was visualized on glass smears under light microscopy (Magnification 40×).

intracellular bacteria was noted upon addition of the cells pre-incubated with the anti-r*Vp*FimH antisera (7.65×10^5 CFU/mL and 1.675×10^6 CFU/mL with 1:50 and 1:100 dilution of the antisera) in comparison to that pre-incubated with pre-immune serum (8.14×10^6 CFU/mL at 1:50 dilution). The decrease in the population of adhered and intracellular bacterial count was inversely proportional to the dilution of the anti-r*Vp*FimH antiserum.

3.5. Protective efficacy of rVpFimH immunization against V. parahaemolyticus

Passive challenge of normal (unimmunized) mice with $2\times LD_{50}$ dose of *V. parahaemolyticus* pre-incubated with neat anti-r*Vp*FimH antiserum showed 100 % survival for the experimental period. The mice remained healthy even beyond the observation period of 10 days. On the other hand, all the mice challenged with $2\times LD_{50}$ dose of *V. parahaemolyticus* cells, pre-incubated with the pre-immune serum died within 24 h (Fig. 10A).

In vivo challenge in the r*Vp*FimH-immunized BALB/c mice also

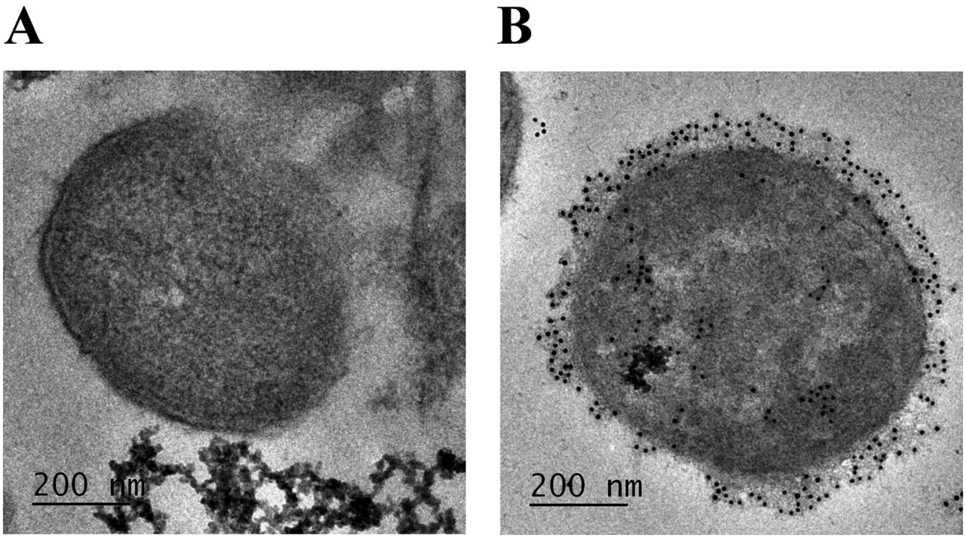

Fig. 7. TEM analysis to detect surface exposed FimH on *V. parahaemolyticus* cells by immunoelectron microscopy using anti-r*Vp*FimH antibodies. TEM images of the immunogold labeled (employing colloidal gold-labelled anti-mouse antibody) mid-log phase *V. parahaemolyticus* cells treated with pre-immune (A) and anti-r*Vp*FimH antisera (B) are shown. Detection of black dots on the fimbria present on the bacterial cell surface only in the cells incubated with the anti-r*Vp*FimH antisera indicate the specificity of the cross-reactivity of the antibodies present in the anti-r*Vp*FimH antisera with the native FimH.

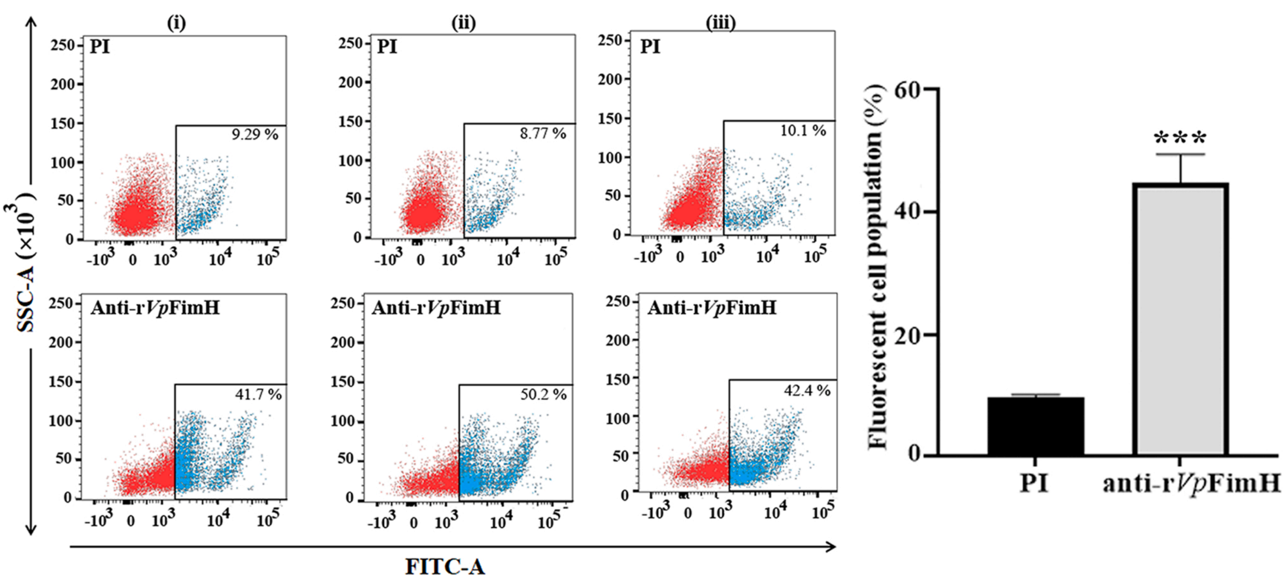

Fig. 8. Cross-reactivity analysis of the anti-r*Vp*FimH antibodies with the *V. parahaemolyticus* FimH using FACS. *V. parahaemolyticus* (MTCC#451) cells incubated with either pre-immune (PI) or anti-r*Vp*FimH antisera (1:50) (each in triplicate) were stained with FITC-conjugated secondary antibody (1:200 in 1 × PBS). The scatter plot shows the percentage of FITC-stained fluorescent cell population of three independent experiments. The bar diagram on the right shows mean ± SD of fluorescent cell population of the three independent experiments, shown on the left. *, $p \leq 0.001$ with respect to cells treated with PI.

showed 100 % survival for 10 days when challenged with 2×LD_{50} CFUs of *V. parahaemolyticus.* The challenged mice were found to be healthy, and no mortality was noticed even beyond 10 days (Fig. 10B). All the control PBS-immunized mice died within 12 h of the bacterial challenge.

4. Discussion

Continued use of antibiotics has led to the development of multi-drug resistant *V. parahaemolyticus*, the major *Vibrio* species responsible for vibriosis in humans. Due to this, the mariculture industry suffers major losses as significant percentage of cargos are rejected for the fear of transmission of bacteria from the infected marine species to humans. Due to the development of drug resistance, prophylactic measures such as vaccination remains the method of choice to control *V. parahaemolyticus* infection. A number of surface proteins have been reported to be crucial for primary interaction and adhesion of the bacterium to the host cell, and therefore can serve as effective vaccine candidates. Of these, type 1 fimbrial adhesins play a major role in bacterial adhesion and colonization in the host gut, and significantly contribute to the virulence of several bacteria (Connell et al., 1996; Klemm and Schembri, 2004; Pereira et al., 2015; Choi et al., 2016; Guevara et al., 2016; Asadi et al., 2018). Hence, these adhesins are effective antigens and few of these have also been reported to generate protective immunity. However, FimH has neither been characterized in *V. parahaemolyticus*, nor has been evaluated for its potential as a subunit vaccine candidate against the bacterium.

In the present study, we have characterized the FimH of *V. parahaemolyticus*, assessed the immunogenic and vaccine potential of r*Vp*FimH in BALB/c mice. The presence of type 1 fimbrial component in *V. parahaemolyticus* was confirmed and the nucleotide sequence of *V. parahaemolyticus* FimH (*Vp*FimH) identified showed 84 % identity with *E. coli* FimH, one of the most investigated FimH. Like FimH of the other bacteria, presence of lectin binding domain at the N-terminus of *Vp*FimH was evident. The evolutionarily relationship of *Vp*FimH with other bacterial FimH by multiple sequence alignment (MSA) and the phylogenetic tree generated from the MSA revealed varying degree of

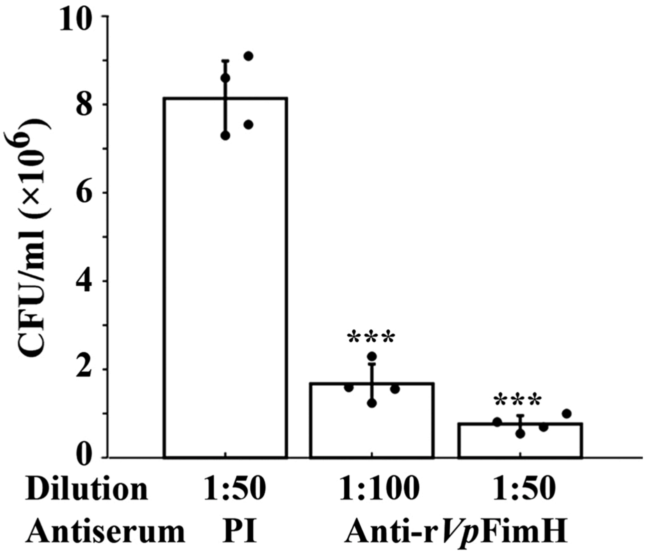

Fig. 9. *In vitro* **e**valuation of neutralization potential of anti-r*Vp*FimH antiserum. Adhesion and invasion of *V. parahaemolyticus* (1×10^7 CFU) pre-incubated with either pre-immune serum (PI,1:50; 1 mL 1 × PBS) or anti-r*Vp*FimH antiserum (1:50 and 1:100 in 1 mL 1 × PBS) for 2 h at 37 °C on HCT116 cells (1×10^5 cells/1000 μl/well; multiplicity of infection, 1:100) was analysed. The treated cells were incubated for 2 h at 37 °C in 5 % CO_2 atmosphere. Microdilution method was used to determine the number of adhered and intracellular *V. parahaemolyticus* cells. Bee-Swarm plot of CFU/mL of four replicates of each experimental sample is shown. The significance (*p* value) level is determined with respect to bacterial cells pre-incubated with PI. ***, $p \leq 0.001$).

the percentage identity of the *Vp*FimH with other bacterial FimH, ranging between 35 % to > 99.67 %. Interestingly the phylogenetic analysis of FimH with other bacteria revealed that the *Vp*FimH clustered with the FimH of few *Citrobacter* species, which is in agreement with earlier report putforth by Letchumanan et al. (2016), which showed that the genome sequence VP103 showed greater phylogenomic affiliation to *Citrobacter* instead of other *Vibrio* species.

Analysis of genomes of different *V. parahaemolyticus* strains available in the GenBank showed that only two strains (VP152 and VP103) have annotated and fully characterized FimH sequence (KKY42922.1 and KKF71749.1, respectively) in their genome, which showed 100 % identity. Thus, it is likely once the genomes of different *V. parahaemolyticus* are annotated, the FimH sequences are going to share significant similarity, thus providing global utility to the *V. parahaemolyticus* FimH as prototype vaccine. Also, intraspecific and interspecific gene transfer through transformation, transduction and conjugation among different bacteria are a way of life and an adaptive virulence strategy (Gyles and Boerlin, 2014; Kerr et al., 2014). Fimbrial alleles have been reported to be associated with the adaptive virulence strategy in different bacteria including *V. parahaemolyticus* and other *Vibrio* species (Silberger, 2008; Madsen et al., 2012; Deng et al., 2019; Muthukrishnan et al., 2019). Lateral gene transfer of pandemic *V. parahaemolyticus* virulence gene to avirulent *V. parahaemolyticus* has been reported to turn the avirulent strain to virulent one (Silberger, 2008; Muthukrishnan et al., 2019; Espejo et al., 2017; Fu et al., 2020). Therefore, the confirmed presence of FimH in *V. parahaemolyticus* strains VP152 and VP103 highlights the possibility of transfer of this gene to other *V. parahaemolyticus* strains, in which it may not be present currently. Hence, the putative FimH vaccine would have the potential application against not only *V. parahaemolyticus*, but other bacteria as well.

For assessing the immunogenic and vaccine potential of *Vp*FimH, it was necessary to produce large amounts of r*Vp*FimH. Purification of recombinant functional adhesins poses several difficulties due to their tendency to aggregate, and susceptibility to proteolytic degradation. This hinders production of recombinant FimH required in large quantities for assessing the vaccine potential (Hultgren et al., 1989). Attempts have been made to stabilize the *E. coli* FimH by co-expression with chaperone FimC, which was purified from the periplasmic fraction, and also by expressing it in translation fusion with fliC. However, these resulted in relatively lower yields of recombinant proteins (20 mg/L and 3 mg/L, respectively (Vetsch et al., 2002; Asadi et al., 2012). In the present study, the r*Vp*FimH expressed as inclusion bodies. The misfolded aggregates of the r*Vp*FimH were solubilized and the cellular impurities adsorbed on the hydrophobic inclusion bodies were removed by repeated washing with detergent (2 % sodium deoxycholate, w/v) and buffer containing relatively lower concentration (2 M) of urea. Subsequently, the r*Vp*FimH was purified from the inclusion bodies solubilized with higher urea concentrations employing metal affinity chromatography. As the refolding of denatured proteins purified from the inclusion bodies to biologically active conformation is cumbersome and often despite high yield of recombinant protein produced as inclusion bodies, biologically active stable proteins are not suitable for therapeutics and industrial applications. Solubilization in mild detergent followed by pulse refolding has been reported to aid in the production of a number of biologically active proteins (Mahmoodi et al., 2017; Kötzler et al., 2017). In the present study, the purified r*Vp*FimH was successfully solubilized using pulsatile renaturation method (Mahmoodi et al., 2017) by adding r*Vp*FimH dropwise in osmolytic buffer (5 % sucrose) at the flow rate of 0.1 mL/min. Vetsch and his associates have demonstrated refolding of the recombinant FimH, and its domains (lectin and pilin) *in vitro* using low protein concentrations, low ionic strength and rapid dilution method (Vetsch et al., 2002). Their studies demonstrated that while the lectin domain of FimH folded to its native form *in vitro* and was stable, the refolded pilin domain was thermodynamically unstable and

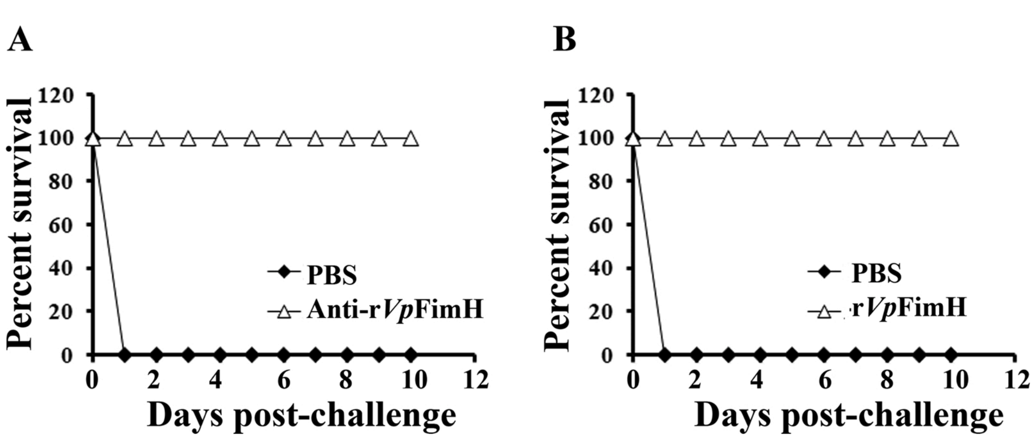

Fig. 10. Evaluation of protective immunity by bacterial challenge assay.**(A)** Passive challenge of naïve mice (n = 10 per group) was performed with $2\times LD_{50}$ (5×10^8 CFUs in 50 μL 1 × PBS, IP) of live *V. parahaemolyticus* (MTCC#451) cells pre-incubated with equal volume of either the pre-immune (PI) or anti-r*Vp*FimH antisera. **(B)** Protective efficacy of rVpFimH immunization in BALB/c mice. The r*Vp*FimH-immunized and PBS-immunized mice (n = 10 each group)mice were challenged with $2\times LD_{50}$ dose of *V. parahaemolyticus* (MTCC#451) cells and monitored for 10 days post-challenge.

remains in a mixture of unfolded and folded state and was prone to aggregation. Since various sugar osmolytes including sucrose have been reported to inhibit non-specific protein aggregation as these molecules interfere with intermolecular hydrophobic interactions, we used refolding buffer containing sucrose (Chen et al., 2015 and references therein) and dialyzed in a buffer cotnaing 5 % glycerol as it has also been reported to stabilize the proteins and prevent aggregation (Mahmoodi et al., 2017 and references therein). Addition of 5 % glycerol prevented aggregation of r*Vp*FimH during long term storage and was maintained in the soluble form. Employing this strategy, we could obtain ~43.2 mg/l of the refolded r*Vp*FimH at the shake flask level, that was used for immunization studies.

Vaccine studies using full length type 1 fimbrial component of *E. coli* failed to generate protective immune response, and even at a minimal dilution of 1:50, the antisera could not completely block bacterial adhesion and infection (Langermann et al., 1997). On the other hand, recombinant FimH, fimbrial tip protein of uropathogenic *E. coli,* conferred protection against recurring urinary tract infection (Langermann et al., 1997). The major subunit of type 1 fimbria *i.e.* FimA, is composed of only pilin region, has been reported to give significantly lower protection (<50 %) (Musa et al., 2014), than the lectin domain (~94 %) (Langermann et al., 1997). This could be due to the fact that the the lectin domain folds to its native conformation, and the availability of both conformational and linear epitopes makes the antigenic response generated against the protein more effective and protective, whereas the pilin domain is prone to aggregation (Vetsch et al., 2002). In line with the reports on *E. coli* FimH, the immunization of BALB/c mice with the r*Vp*FimH emulsified with alumunium hydroxide gel generated an effective immunogenic response evident from r*Vp*FimH specific high antibody titers (end point titers, 1:1,000,000) of the anti-r*Vp*FimH antiserum. Long lasting protective immune response has been attributed to, in addition to the antigen, the use of a safe and stable adjuvant (Batista-Duharte et al., 2018). We used alum as it is an approved adjuvant for several licensed vaccines for human use against a variety of infections (Di Pasquale et al., 2015). The r*Vp*FimH adjuvanated with alum generated robust immune response that is observed with other adhesins, OMPs and fimbrial proteins adjuvanated with Complete Freund's adjuvant/ incomplete Freund's adjuvant, established immunopotentiator known to generate very high immune response (Coffman et al., 2010). The end point titer of > ~1:1,000,000 of the anti-r*Vp*FimH antiserum is comparable with that obtained upon immunization with the fimbrial proteins of other bacteria (Langermann et al., 1997; Liu et al., 2009; Yang et al., 2011; Musa et al., 2014; Chalton et al., 2019). Immunization with the *E. coli* FimH in fusion with FimC that gave high antibody titers showed significant decline (100–150 fold) in bacterial populations upon active challenge of the immunized animals (Langermann et al., 1997; Langermann and Ballou, 2001).

Significantly higher levels of IgG1, IgG2a and IgG2b in the antisera suggest generation of mixed immune response against the r*Vp*FimH. IgG1 levels (indicator of Th2- mediated immune response) remained significantly higher than IgG2a and IgG2b on all the days, and the ratio of IgG1/IgG2a and IgG1/IgG2b remained constantly greater than 1, clearly demonstrating a Th2-biased mixed immune response. A significant increase in IgA levels in the anti-r*Vp*FimH antisera is in agreement with earlier reports wherein enhanced level of IgA post-intranasal immunization with *E. coli* FimH and lectin domain of *E. coli* FimH have been observed (Poggio et al., 2006). Increased IgG levels against the r*Vp*FimH suggest robust immune response and are likely to confer protection against *V. parahaemolyticus*, as immunization with *E. coli* FimH in fusion with FliC that enhanced IgG production resulted in restricted colonization of uropathogenic *E. coli* in murine model (Asadi et al., 2013).

Bacterial agglutination due to interaction of the bivalent antibodies present in the serum with multiple epitopes of the target antigen on bacterial cells result in visible clumping and prepare them for phagocytosis plays a key role in bacterial clearance from the organism (Murphey et al., 1979; Reed et al., 1983; Kind, 2010). Bacterial agglutination by antibodies have been reported to inhibit bacterial colonization in the gut (Roche et al., 2015; Mitsi et al., 2017). Therefore, it was important to establish if the anti-r*Vp*FimH antisera has agglutinating antibodies. The ability of the anti-r*Vp*FimH antisera to agglutinate the live *V. parahaemolyticus* cells confirm specific interaction of the antigen present on the surface of *V. parahaemolyticus* cells with the specific anti-FimH antibodies present in the anti-r*Vp*FimH antisera. With increasing dilution of the antisera, decreased agglutination of *V. parahaemolyticus* was observed due to the disturbed proportions of antigen or antibody, called as postzone or prozone effect. Inhibition of agglutination of the *V. parahaemolyticus* cells by the anti-r*Vp*FimH antibodies by mannose clearly confirmed the interaction of the anti--r*Vp*FimH antibodies with the lectin binding domain of the *Vp*FimH on its cell surface. Such an interaction is of importance as this would inhibit *V. parahaemolyticus* binding to the host cells. The specificity of this interaction is further established by absence of any agglutination when the anti-r*Vp*FimH antisera incubated with r*Vp*FimH prior to addition to the bacterial cells.

Flow cytometry analysis was also performed to check if the antibodies present in the anti-FimH antisera were functional and could cross react with the FimH on *V. parahaemolyticus* cell surface (Kisiela et al., 2006; Fabbrini et al., 2012). Using flow cytometry, it has been established that agglutinating antibodies raised against outer membrane protein of *Aeromonas hydrophila* specifically bind to the antigen on the *A. hydrophila* cell surface (Sharma and Dixit, 2015; 2018). Using the same strategy, growth inhibitory antibodies have been identified against BamA of *E. coli* (Vij et al., 2018). We used side scatter gating to identify cells that have increased complexity (granularity) due to binding of the anti-FimH antibodies to the FimH on *V. parahaemolyticus* cells. Incubation of *V. parahaemolyticus* cells incubated with the control (pre-immune sera) showed very less percentage of fluorescent cells population evident from limited side scatter, as this antisera did not contain antibodies that could interact with any protein on the bacterial cell surface. Using the same gating, significantly enhanced fluorescent cell population of the bacteria (indicated by higher side scatter) are observed when *V. parahaemolyticus* cells were incubated with anti-r*Vp*FimH antisera followed by incubation with FITC conjugated secondary antibodies, in comparison to the cells incubated with pre-immune sera. This is because the specific anti-FimH antibodies present in the antisera are able to interact with FimH present on the *V. parahaemolyticus* cell surface and the the stained cells emit higher fluorescence intensity resulting in higher side scatter. These results are also corroborated with immunogold labeling analysis using anti-r*Vp*FimH antisera.

Since agglutinating antibodies have been reported to inhibit colonization of *Streptococcus pneumoniae* in mouse model (Roche et al., 2015). FimH binds to the mannosylated residues on the host cells through its lectin binding domain (Sauer et al., 2016), therefore the agglutinating antibodies present in the anti-r*Vp*FimH antisera would be expected to specifically bind to the FimH on bacterial cell surface and inhibit adhesion of *V. parahaemolyticus* cells to the host cells. The specific interaction of the anti-r*Vp*FimH antibodies with the FimH on bacterial cell surface and subsequent neutralization of the bacterial adhesion and invasion is evident with significantly reduced number of adhered and intracellular *V. parahaemolyticus* cells in HCT116 cells *in vitro* when the bacteria were pre-incubated with the anti-r*Vp*FimH antiserum prior to addition to the host cells. These data clearly suggest that the anti-r*Vp*FimH antibodies were able to mask the lectin binding domain of the FimH present on the *V. parahaemolyticus* cell surface thus blocking its interaction with the mannosylated oligosaccharide residues present on the colorectal carcinoma HCT116 cells. Our results are in agreement with the earlier reports putforth by other investigators who reported prevention of mucosal *E. coli* infection upon immunization with FimH (Langermann et al., 1997; Luna-Pineda et al., 2016). These data further demonstrate the potential of the anti-r*Vp*FimH antiserum in deterring biofilm and colonization mediated by *V. parahaemolyticus.*

Modulation of cytotoxic and humoral immune response is governed by the plethora of cytokines secreted by antigen presenting cells and other plasma cells (Mukovozov et al., 2008). Proliferation of both the immunological memory B and T cells is crucial for eliminating the pathogen and conferring long lasting immune response (Crotty et al., 2003). In the present study, the r*Vp*FimH-sensitized lymphocytes exhibited significantly augmented proliferation index clearly indicating generation of T cell memory response.

Generation of a balanced immune response that lead to production of specific and long term protective antibodies is highly desirable characteristic of a vaccine candidate (Spellberg and Edwards, 2001). In line with antibody isotyping data that suggested a Th2-biased mixed immune response, increased levels of both IFN-γ and IL-4 in the culture supernatant of the r*Vp*FimH-primed splenocytes, stimulated with r*Vp*FimH, confirmed mixed immune response. The immediate immune response generated against the r*Vp*FimH is similar to that observed with *V. parahaemolyticus* infections in murine models and humans, where in an increase in proinflammatory cytokines including IFN-γ was observed (Qadri et al., 2003; Waters et al., 2013). It is noted that at 72 h, there is a shift from Th1 (indicated by decline in IFN-γ) towards Th2 (indicated by rise in IL-4 levels) immune response. Such a switch from cell mediated response to humoral helps the organism from the adverse effects caused by overwhelming persistent cytotoxic immue response (Livingston et al., 1999). Global cytokine array analysis of the culture supernatant collected at 72 h also confirmed that r*Vp*FimH was able to stimulate all the three arms (Th1/Th2/Th17) of immune response. The cytokines/chemokines with enhanced levels secreted by CD4+ Th1 cells included IP-10, IL-3, TNF-α, KC, IL-6, GM-CSF, JE, MIG, sICAM-1, CCL3, IFN-γ, CCL4, IL-1α, CXCL2, IL-17, IL-1β, CCL5, IL-1ra, IL-2, and by CD4+ Th2 cells included IL-3, IL-4, TNF-α, IL-5, G-CSF, M-CSF, IL-6, GM-CSF, IL-7, I-309, IL-10, IL-13. In line with our observations, other outer membrane proteins of *V. parahaemolyticus* and other *Vibrio* species have been reported to induce both pro- and anti-inflammatory cytokines, and activate both the arms (innate and adaptive) immune responses (Yang et al., 2019). Our results are in agreement with the earlier reports wherein an increase in TNF-α and IL-6 levels in monocytes upon immunization with a recombinant outer membrane protein OmpU of *V. parahaemolyticus* (Gulati et al., 2019) and in IL-1β, IL-8, IL-10 and CCL4 was observed upon immunization with the outer membrane protein OmpK of another *Vibrio* species, *V. alginolyticus* (Silvaraj et al., 2020). An increase in IL-17 levels that is secreted by both CD4+ Th1 and CD4+ Th17 was also noted. Similar increase in the levels of TNF-α, IL-1α, IL-1β, IL-6, IL-10, IL-12, IL-23, and IFN-γ has been reported earlier in human intestinal mucosa and murine mucosa or macrophage earlier upon *V. parahaemolyticus* infection in humans (Qadri et al., 2003; Waters et al., 2013). These investigators also reported an increase in neutrophil attracting chemokines CXCL1 and CXCL2 in response to *V. parahaemolyticus* infection in mice (Yang et al., 2019). The stimulation of inflammatory cytokines activate cascade of cytokines, chemokines production, T cell differentiation and recruitment of different immune cells against pathogens, for their effective clearance from the host (Waters et al., 2013 and references therein). The maximum secretion of Th1 related chemoattractant from CC and CXC chemokine family: CCL3 (MIP1α/Macrophage Inflammatory Protein 1α), CCL4 (MIP1β), and CCL5 (RANTES/ regulated upon activation, normal T-cell expressed, and secreted), CXCL2 (MIP-2) was observed. These chemokines are secondary pro-inflammatory mediator secreted from macrophages and are activated by primary pro-inflammatory moderator. An increase in CCL3 (MIP1α) levels has been reported to be associated with protective immunity against intracellular pathogen like *L. monocytogenes* (Gregory et al., 2001). Elevated CCL3 (MIP1α) and CCL4 (MIP1β), related to inflammatory infection, trigger the secretion of other proinflammatory cytokines namely TNF-α, IL-6 and IL-1β from macrophages. Elevated levels of TNF-α have been reported to clear intracellular infection of *Francisella tularensis* by stimulation of nitric oxide and memory response (Cowley et al., 2007). Significantly elevated CCL5 (RANTES) levels correlate well with the observed increase in large numbers of Th1 cytokines in the culture supernatant, and serum IgG2a and IgA levels. The increased level of TNF-α and subsequently released Th1 cytokines and pro-inflammatory chemokines have been associated with protection against a variety of infections (Cowley et al., 2007; Tavares et al., 2017). These Th1 chemokines have been reported to assist in chemotaxis by releasing granulozymes, ROS and thereby inhibit spread of infection (Tavares et al., 2017). In addition to the stimulation of Th1 and Th2 immune response, a significant increase in IL-17 from the newly discovered third T cell subset Th17 was also noted in the culture supernatant of r*Vp*FimH-primed and stimulated splenocytes. This is of immense significance as IL-17 plays a significant role in confering protection against both intracellular as well as extracellular pathogen.While the Th2 and Th1 play a mutual antagonistic role in conferring a balanced immune response, the Th17 cells help in secretion of IFN-γ that result in switching of Th17 phenotype to Th1. Thus an interplay of all the three arms activated by r*Vp*FimH is likely to help maintain an optimal protective immune response against *V. parahaemolyticus*.

Thus, like other fimbrial proteins of other bacteria (Liu et al., 2009; Yang et al., 2011; Asadi et al., 2013; Musa et al., 2014; Asadi et al., 2018), the r*Vp*FimH was able to stimulate both humoral and adaptive immune response, and is therefore expected to confer effective protection.

One hundred percent survival of mice in passive challenge studies of the normal unimmunized mice with $2\times LD_{50}$ CFUs of *V. parahaemolyticus* pre-incubated with the anti-r*Vp*FimH antisera validated the *in vitro* data on agglutination, and adherence and invasion analysis using HCT116 cells. Further 100 % protection of r*Vp*FimH-immunized BALB/c mice with the lethal dose of *V. parahaemolyticus* confirmed generation of protective immunity by the r*Vp*FimH. The observed protection can be correlated with stimulation of long term memory response and generation of both humoral and cellular immune response, a highly desirable trait of a vaccine candidate. This possibly inhibited primarily adhesion of the bacteria to the host cells (evident from reduced count of adhered and intracellular *V. parahaemolyticus* cells pre-incubated with the anti-r*Vp*FimH antisera to HCT116 cells). Similar results have been obtained with recombinant full length FimH of *E. coli* alone, its fragments or FimH in fusion with other proteins (FimH-CsgA and FimH-CsgA-PapG) that activated cellular immune response and reduced colonization of *E. coli* in mice (Langermann et al., 1997; Luna-Pineda et al., 2016). It has been observed with many *V. parahaemolyticus* OMPs/fimbrial adhesins that generate effective immune response, but fail to protect the animals against the bacterial challenge. In the present study, immunization with the r*Vp*FimH elicited strong protective immune response providing 100 % protection against the bacterial challenge which is far better than that observed (~30 % - ~75 % on immunization with other surface proteins/fimbrial adhesins or outer membrane proteins of *V. parahaemolyticus* both in murine model as well as fish model (Mao et al., 2007; Li et al., 2014; Peng et al., 2016). Likewise, relatively lower protection with fimbrial proteins of other bacteria namely F41 of *E. coli* (80 %), FimA, of *Salmonella enterica* (50 %) have been reported (Liu et al., 2009; Musa et al., 2014). Thus, FimH of *V. parahaemolyticus* appears to be more effective in conferring protection than these fimbrial proteins. However, it is to be noted that the bacterial adhesins including type 1 fimbriae exhibit tropism, and alter their conformation in response to external/environmental stimuli (Tchesnokova et al., 2011; Kalas et al., 2017). Thus, while the fimbrial proteins are able to generate an effective immune response, not necessarily the antibodies are able to effectively neutralize the fimbrial attachment to the host. However, use of a polyclonal vaccine may overcome this obstacle as being generated against the whole protein, the antibodies will be able to recognize multiple epitopes (conformational and linear), which could be displayed in different conformations, and thus inhibit the bacterial attachment to the cells.

To summarize, we purified the extracellular minor type 1 fimbrial tip adhesin, FimH from the chaperone-usher pathway of *V. parahaemolyticus* from the insoluble fraction and successfully refolded through pulse refolding. Thus, the r*Vp*FimH immunization generated a potent mixed immune response and neutralizing antibodies in BALB/c mice against *V. parahaemolyticus.* Further, activation of all the three arms of immune response and generated long lasting protective immunity by the r*Vp*FimH clearly demonstrated its potential as a subunit vaccine against *V. parahaemolyticus* in murine model and can be further evaluated for control of vibriosis.

CRediT authorship contribution statement

Aparna Dixit, **Lalit C. Garg**, **Devapriya Choudhury:** Conception and design of the study.

Sweta Karan: Execution of experiment and acquisition of data.

Sweta Karan, **Lalit C. Garg**, **Aparna Dixit:** Analysis and/or interpretation of data.

Sweta Karan, **Aparna Dixit:** Preparation of the original manuscript draft.

Sweta Karan, **Lalit C. Garg**, **Aparna Dixit**, **Devapriya Choudhury:** review & editing of the manuscript and approval of the final version of the manuscript.

Aparna Dixit, **Devapriya Choudhury**, **Lalit C. Garg:** Supervision and project administration.

Funding

This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

Ethical statement

The study does not involve human subjects. Animal studies were approved by the Institutional Animal Ethics Committee (IAEC) of the Jawaharlal Nehru University, New Delhi and all animal procedures were performed in compliance with the IAEC guidelines.

Declaration of Competing Interest

The authors declare no conflict of interest.

Acknowledgements

The authors acknowledge intramural support from the Jawaharlal Nehru University, New Delhi. SK thanks the Indian Council of Medical Research, New Delhi for research fellowship.

Appendix A. Supplementary data

Supplementary material related to this article can be found, in the online version, at doi:https://doi.org/10.1016/j.molimm.2021.05.005.

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